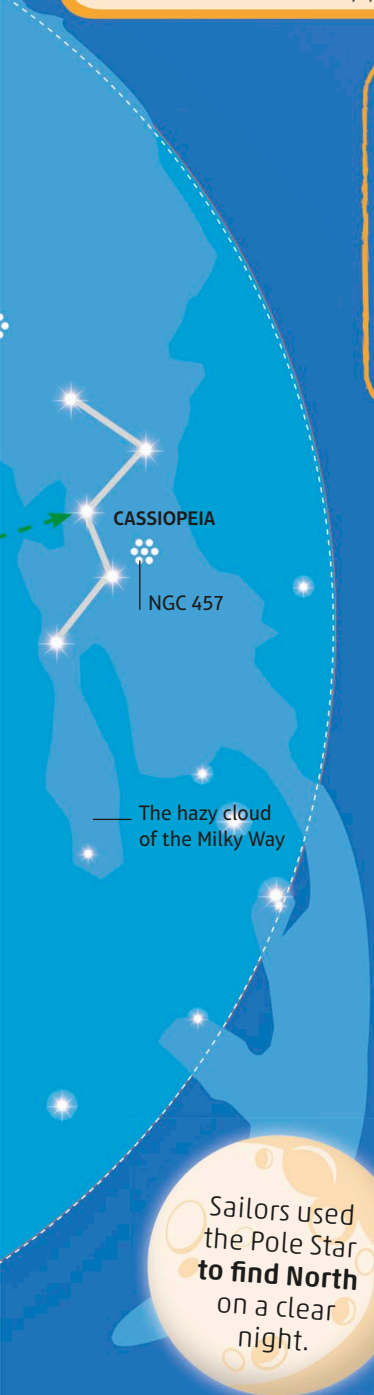


STARGAZING BASICS

Northern star hopping

To find your way around the night sky, it is easiest to pick out some obvious landmarks and then trace imaginary lines from these to other constellations. This technique is called star hopping and is easy to do with the naked eye, though you can see more if you have binoculars. The chart on this page shows how to star-hop around Polaris, the Pole Star, which is visible to people who live in Earth's northern hemisphere.



FINDING THE WAY

- 1 Find the Plough (in red) and locate the two stars furthest from its "handle", which are known as the Pointers. Draw an imaginary line (purple) through the Pointers and extend it to a star of similar brightness. This is the Pole Star and it is always due north.
- 2 Extend the line from the Pointers past the Pole Star to reach the constellation Cepheus, which looks a bit like a lopsided house. Binoculars will reveal the bright red Garnet Star at the base of the house – the reddest star visible to the naked eye.
- 3 Now draw a line (green) from the third star down the Plough's handle through the Pole Star to find the constellation Cassiopeia, which looks like a flattened "W". If you have binoculars, look for a star cluster just below the central peak of the "W".
- 4 The large but faint constellation Draco (the Dragon) is best seen under very dark skies. A line (pink) from the fourth star down the Plough's handle cuts across the dragon's body and carries on to its head, a pattern of stars called the Lozenge.
- 5 Track down one of the brightest galaxies in the sky with binoculars: follow a diagonal line (orange) across the Plough's rectangle of stars, and continue in the same direction looking for a pair of tiny fuzzy patches. These are the galaxies M81 and its fainter companion M82.

